ZENBUY: SMART WISHLIST & AI SHOPPING GUARDIAN

Product Specification, Behavioral Economics Framework, Technical Architecture & Monetization Blueprint

1. Executive Summary & Product Vision

ZenBuy is an ambient, AI-powered browser companion and smart wishlist engineered to curb impulsive online consumer spending. Rather than operating as a blunt, punitive website blocker—which empirical market data proves users quickly uninstall or bypass—ZenBuy leverages behavioral economics to introduce **adaptive "good friction"**, **cognitive cooling-off periods (24–48 hours)**, and **cross-store value optimization**.

By combining **Jev AI** (TypeSafe AI's sub-50ms System-1 structured decision engine) and **Google Gemini 3 Flash** (a System-2 Socratic reasoning model), ZenBuy silently differentiates between necessary household staples (e.g., groceries, diapers, replacement cables) and high-risk hedonic impulses. When an impulse purchase is detected, ZenBuy intercepts the checkout urge and channels the consumer's acquisition dopamine into an automated "Quarantine Vault."

> TIP
>
> **Core Commercial Breakthrough:** ZenBuy is 100% free to consumers. It monetizes not through prohibitive SaaS paywalls, but via an **Affiliate Optimization Model**. When a consumer deliberately decides to proceed with a purchase after their 48-hour reflection period, ZenBuy searches across major US retailers to find the lowest price, coupon code, or cash-back deal, earning a 3%–10% referral commission on deliberate, non-regretted transactions.

2. Empirical Problem Analysis: Impulsive Buying in the USA

Impulsive consumer spending in the United States represents an escalating systemic crisis catalyzed by modern frictionless e-commerce architecture, one-click checkouts, algorithmic social commerce (TikTok Shop, Instagram), and decoupled Buy Now, Pay Later (BNPL) credit models.

2.1 Key Statistics & Market Data (2026 Benchmarks)

- **Prevalence: 81% to 91%** of American consumers admit to making frequent, unplanned impulse purchases.
- **Average Expenditure:** The average US consumer spends between **$282 and $318 per month** on unplanned purchases, totaling **$3,400 to $3,800 annually per adult**.
- **Transaction Frequency:** Shoppers make an average of **9.7 to 11.3 unplanned purchases monthly—**nearly one impulse buy every 72 hours.
- **High-Ticket Creep: 27%** of consumers report having made a single unplanned impulse purchase exceeding $1,000.
- **Buyer's Remorse & Financial Distress: 62%** of consumers report experiencing immediate regret after unplanned buys, and **34%** report direct debt, financial strain, and anxiety directly attributable to impulse spending.

2.2 Digital Catalysts Fueling the Problem

1. **Frictionless Checkout Acceleration:** Amazon 1-Click, Apple Pay, and browser autofill have compressed the window between product discovery and financial commitment to under 8 seconds.
2. **Decoupled Payment Micro-Installments:** BNPL providers (Klarna, Afterpay, Affirm) fragment $120 purchases into "4 easy payments of $30," numbing the consumer's perception of debt.
3. **Social Commerce & Algorithmic Targeting:** Platforms like TikTok Shop and Instagram utilize hyper-personalized feeds during passive emotional scrolling, driving an average of 14.7 unplanned purchases per user monthly.

3. Behavioral Economics Foundations

ZenBuy's intervention mechanics are directly grounded in academic behavioral economics and cognitive bias research (referencing frameworks cataloged by *The Decision Lab*, Daniel Kahneman, Amos Tversky, Drazen Prelec, and George Loewenstein):

Cognitive Bias / Heuristic	How Retailers Exploit It	ZenBuy Counter-Measure
Present Bias & Hyperbolic Discounting	1-Click checkout delivers immediate dopamine hit; financial consequences are delayed.	**48-Hour Cart Quarantine:** Enforces a cooling-off window; dopamine levels normalize within 24–48 hours, dissipating up to 70% of impulse urges.
Pain of Paying & Decoupling *(Prelec & Loewenstein)*	Digital wallets and BNPL remove the visceral sting of parting with physical cash.	**Wage & Debt Salience:** Re-couples spending by translating dollars into "Hours of Labor" (e.g., "$120 = 5.2 Hours of Work") and unmasking BNPL obligations.
Scarcity Heuristic & Loss Aversion *(Kahneman & Tversky)*	Fake urgency ("Only 2 left!", "Deal expires in 04:32") triggers FOMO.	**Dark Pattern Stripper:** Automatically hides or blurs artificial countdown timers, fake stock counters, and strikethrough price anchors.
Zero-Price Effect & Pseudo-Savings *(Dan Ariely)*	"Add $18 more for Free Shipping!" tricks consumers into spending $25 on junk to save $6.	**Net Shipping Loss Calculator:** Explicitly calculates: "You are spending $18 to save $6 on shipping. Net loss: -$12."
Ego Depletion & System 1 Dominance *(Dual-Process Theory)*	Shoppers browsing late at night (10 PM–4 AM) suffer cognitive fatigue, disabling rational control.	**Late-Night Shield:** Automatically locks checkouts between 10 PM and 5 AM, requiring overnight sleep before purchase verification.
Endowment Effect & Mental Ownership	Adding items to cart establishes psychological ownership before purchase.	**Dopamine Vault Substitution:** Saving the item to the ZenBuy Vault provides the mental satisfaction of "capturing" the item without spending money.

4. Product Overview & User Flows

4.1 Desktop Experience (Chrome, Edge, Brave Web Extension)

1. **Silent Background Evaluation:** When browsing an e-commerce page (Amazon, Walmart, Target, Shopify stores), ZenBuy extracts product metadata. Jev AI evaluates the page in under 50ms.
2. **Adaptive Button Injection:**
    - *For Staples (Toilet paper, medicine):* Zero interference. Normal checkout buttons remain.

- *For Discretionary Wants:* Injects a secondary `🧊 Put on Ice (48h)` button beside "Add to Cart".
- *For High-Risk Impulses:* Intercepts the "Add to Cart" click, opens the ZenBuy reflection modal, and presents the labor-cost trade-off.

3. **The 48-Hour Decision Card:** When the timer expires, the user receives an alert displaying price history, cross-store comparison, and two primary actions: **[Archive & Save]** or **[Unlock Best Price Checkout]**.

4.2 Mobile Experience (iOS & Android Architecture)

> IMPORTANT
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> **Solving the Mobile OS Sandboxing Constraint:** Native apps (Amazon App, TikTok App) operate in isolated sandboxes and strictly prohibit 3rd-party code injection. ZenBuy solves this through three platform-compliant mechanisms:

1. **Native Share Sheet Intermediary:** Inside the Amazon or TikTok app, the user taps the native **Share** icon and selects **"ZenBuy: Put on Ice"**. An iOS bottom sheet appears displaying the labor cost and saving the item into the 48-hour vault.
2. **App-Launch Intervention (Apple Screen Time API):** When the user attempts to open the Amazon or TikTok app during vulnerable hours (e.g., after 10 PM), an iOS Shortcut / Screen Time prompt introduces a 6-second pause: *"Are you opening to buy a specific planned item, or browsing?"*
3. **Safari Mobile Web Extension:** Full DOM manipulation on mobile Safari for web-based e-commerce.
4. **Lock Screen Push Notifications (APNs / FCM):** 48 hours later, a rich notification arrives with one-tap actions: Archive to bank the savings or open the optimized purchase link.

5. Dual-Process AI Architecture: Jev AI & Gemini Flash

To achieve sub-50ms responsiveness and minimize API operational costs, ZenBuy mirrors human cognition with a two-tiered AI pipeline:

5.1 Tier 1: Jev AI (System 1 Fast Triage Layer)

Jev AI (TypeSafe AI) executes real-time, strictly typed classification decisions in 30–50ms. It processes JSON payloads without the token-streaming latency or hallucination risks of conversational LLMs.

```
json
// JEV AI Input Payload
{
  "product_title": "Retro Mechanical Keyboard with RGB Backlight",
  "category": "Electronics > Computer Accessories > Keyboards",
  "price": 139.99,
  "user_local_time": "23:15",
  "has_urgency_timer": true
}

// JEV AI Output Judgment
{
  "classification": "HEDONIC_WANT",
  "impulse_risk_score": 91,
  "confidence": 0.97,
  "recommended_action": "PROACTIVE_INTERCEPT",
  "cooldown_hours": 48
}
```

5.2 Tier 2: Google Gemini 3 Flash (System 2 Socratic Reasoning Layer)

Gemini Flash is invoked *only* when Jev AI flags a high-risk purchase and the user attempts to proceed, or when drafting the 48-hour follow-up recommendation card.

> NOTE
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> **Gemini System Prompt & Execution:**
> **Prompt:** *"The user is attempting to purchase a $139.99 mechanical keyboard at 11:15 PM. Hourly wage: $24. Savings goal: 'Pay down $1,500 credit card debt'. Generate a concise, non-judgmental 2-sentence reality check translating the cost into labor hours and opportunity cost. Encourage a 48-hour cool-down."*
> **Gemini Response:** *"This keyboard equals **5.8 hours of your workday**. Because you're working hard to clear your credit card balance, let's freeze this in your vault for 48 hours. If you still want it on Friday, we'll verify the lowest price online."*

6. Cross-Store Price & Deal Optimization Engine

When the 48-hour cooldown concludes, ZenBuy automatically queries the market to present the optimal purchasing path:

1. **Product Barcode Extraction:** The content script scrapes hidden `schema.org/Product` JSON-LD tags to extract exact **UPC, EAN, or GTIN** identifiers.
2. **Cross-Retailer Price Lookup:** Queries the Google Shopping Index (via DataForSEO / ValueSERP API) across 20+ major US stores (Walmart, Target, Best Buy, eBay, B&H).
3. **Amazon Price Drop & Coupon Scanner:** Queries the Keepa API to inspect whether a clip-coupon is active or if the item is at an all-time low.
4. **AI Variant/Condition Verification:** Jev AI filters candidate results to eliminate refurbished, open-box, or mismatched variants.
5. **Affiliate Deep Link Generation:** Automatically converts the destination URL into an affiliate tracking link via Amazon Associates or Skimlinks.

7. Monetization Strategy & Financial Model

7.1 The Affiliate Monetization Mechanism

Unlike competitor extensions that charge $5/month to consumers who are already struggling financially, ZenBuy operates on a **Free-to-Consumer Affiliate Model**:

- **Amazon Associates:** 1% to 4% commission on eligible physical purchases.
- **Skimlinks / Sovrn Commerce:** Universal aggregator network covering 50,000+ merchants (Walmart, Target, Nike, Nordstrom, etc.) paying 3% to 10% commission.

7.2 Monthly Unit Economics (Per Active User)

Line Item	Monthly Volume per User	Unit Cost	Total Cost / Mo
Jev AI Classification	30 page triage calls	$0.0002 / call	$0.006
Gemini 3 Flash Reasoning	15 Socratic cards	$0.0001 / card	$0.002
Price Search API (Google Shopping / Keepa)	10 48-hour deal lookups	$0.0020 / lookup	$0.020
Cloud Infrastructure (Cloudflare Workers, DB)	100 serverless requests	$0.00005 / req	$0.005
Push Notifications (APNs / FCM)	10 push alerts	Free	$0.000
Affiliate Aggregator Integration	Unlimited	Rev-share	$0.000
TOTAL OPERATING COST PER ACTIVE USER / MONTH			**$0.033 / mo**
ESTIMATED AFFILIATE REVENUE PER ACTIVE USER / MONTH (4% on $90 intentional purchases)			**+$3.600 / mo**
NET PROFIT PER ACTIVE USER / MONTH			**+$3.567 / mo (99.1% Margin)**

7.3 Scale Projections

Active User Base	Monthly Tech Cost	Monthly Affiliate Revenue	Annual Run-Rate (ARR Net)	Annual Consumer Savings
1,000 users	$33 / mo	$3,600 / mo	**$42,800**	$2.52 Million
10,000 users	$330 / mo	$36,000 / mo	**$428,000**	$25.2 Million
50,000 users	$1,650 / mo	$180,000 / mo	**$2.14 Million**	$126.0 Million
100,000 users	$3,300 / mo	$360,000 / mo	**$4.28 Million**	$252.0 Million

8. Technical Stack & Implementation Architecture

Layer	Technologies Selected	Key Responsibilities
Browser Extension	JavaScript / TypeScript, Manifest V3, Webpack	DOM mutation observer, wage badge injection, button interception, dark pattern stripping.
iOS / Android Mobile	Swift / Kotlin (or React Native/Flutter), Safari WebExtension	Native Share Sheet Extension, local storage sync, APNs push receiver.
Edge Serverless Backend	Cloudflare Workers / Hono.js or Google Cloud Run	High-throughput, low-latency API gateway, OpenGraph metadata scraper, token security.
AI Decision Engines	Jev AI (TypeSafe AI) + Google Gemini 3 Flash	Real-time intent triage (Jev) and personalized Socratic copy generation (Gemini).
Price & Catalog APIs	DataForSEO Google Shopping API, Keepa API, UPCitemdb	Cross-store live price scraping, historical pricing, coupon discovery.
Database & Cache	Supabase (PostgreSQL) + Cloudflare KV / Upstash Redis	User profile settings, 48h active vault queue, cached viral product lookups.
Affiliate Infrastructure	Amazon Associates API, Skimlinks Merchant API	Dynamic merchant tracking URL redirection and commission reporting.

9. Ethics, Privacy & Trust Architecture

1. **100% Financial Privacy (Zero Local Data Scraping):** Credit card numbers, banking logins, home addresses, and browsing histories are never extracted or stored. Only public product metadata (URL, Title, Price) is transmitted.
2. **The Primary Metric is Money Saved:** The user's main dashboard prominently displays **"Total Dollars Saved & Quarantined"** as the primary KPI. Unlocked purchases are presented neutrally.
3. **Radical Business Model Transparency:** Onboarding explicitly informs users: *"ZenBuy is free because stores pay us a referral fee if you decide to buy after your 48-hour wait. We never pressure you to purchase."*
4. **Zero Fake Scarcity Notifications:** Cooldown notifications never urge users with artificial urgency; they offer neutral, deliberate choice architecture.

10. Implementation Roadmap (MVP to Scale)

- **Milestone 1 (Weeks 1–3): Core Extension MVP**
  - Build Chrome Manifest V3 extension with Amazon & Shopify DOM scrapers.
  - Implement local "Hours of Labor" wage-to-price converter.
  - Connect Jev AI endpoint for sub-50ms necessity classification.
- **Milestone 2 (Weeks 4–6): Vault & Socratic Engine**
  - Build the 48-hour Quarantine Vault and local storage scheduler.
  - Integrate Gemini 3 Flash for context-aware Socratic reflection cards.
  - Implement Dark Pattern Stripper (hiding fake countdown clocks).
- **Milestone 3 (Weeks 7–9): Price Optimization & Affiliate Linker**
  - Integrate DataForSEO Google Shopping API and Keepa API.
  - Implement UPC/GTIN schema parser for 1:1 cross-store matching.
  - Hook up Skimlinks and Amazon Associates dynamic referral links.
- **Milestone 4 (Weeks 10–12): Mobile Expansion**
  - Launch iOS Share Sheet Extension and mobile Safari Web Extension.
  - Set up Apple Push Notifications (APNs) for 48-hour lock screen prompts.
  - Conduct closed beta testing with r/shoppingaddiction and ADHD focus groups.